INCH-POUND

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MILITARY SPECIFICATION

TRUCK, FIXED PLATFORM, ELECTRIC POWERED, PNEUMATIC RUBBER TIRES

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the heavy duty, electric powered, pneumatic rubber tired, fixed platform truck for handling palletized and unpalletized loads over paved areas and in warehouses.

1.2 Classification. Platform trucks shall be defined by the following parameters:

1.2.1 Type. (see 6.2)

a. Type E - Non-spark enclosed

b. Type EE - Spark enclosed

c. Type EX - Explosion-Proof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Ships Parts Control Center,Attn. Management and Engineering Department,Code 0341 P. O. Box 2020, Mechanicsburg, PA 17055-0788 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Capacities. (see 6.2)

a. 2,000 lbs. (908 KG) with a 300 lb (136 KG) maximum drawbar pull.

b. 4,000 lbs. (1816 KG) with a 600 lb (272 KG) maximum drawbar pull.

c. 6,000 lbs. (2724 KG) with a 1200 lb (544 KG) maximum drawbar pull.

d. As specified (see 6.2).

1.2.3 Deck Sizes (see 6.2)

a. 42"x 78" (106 cm x 198 cm)

b. 42"x 95" (106 cm x 241 cm)

c. 48"x 72" (122 cm x 183 cm)

d. 60"x 108" (152 cm x 274 cm)

e. As Specified (see 6.2)

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards and handbooks. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (see 6.2)

SPECIFICATIONS

FEDERAL

GG-P-455	-Plates and Foils, Photographic (Photosensitive Anodized Aluminum)
QQ-P-416	-Plating, Cadmium (Electrodeposited)

MILITARY

MIL-P-514	-Plate, Identification, Instruction and Marking, Blank

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MIL-T-704 -Treatment and Painting of Material

MIL-F-3541 -Fittings, Lubrication

MIL-G-3859 -Grease Gun, Hand, High Pressure, Lever-Operated, Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity)

MIL-M-3971 -Meter, Time Totalizing, Non-Hermetically Sealed, Electrical, General Specification for

MIL-T-17479 -Trailer, Platform, Warehouse, Wood or Metal Platform, 6000 Pound Capacity, Pneumatic Tires, Fifth Wheel Steer

STANDARDS

FEDERAL

FED-STD-595 -Colors

FED-STD-H28 -Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-129 -Marking for Shipment and Storage

MIL-STD-162 -Material Handling Equipment, Preparation for Shipment, Storage, Cyclical Maintenance, Routine Testing and Processing

MIL-STD-209 -Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment

MIL-STD-461 -Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. Department of Labor

29 CFR, Part 1910 - Occupational Safety and Health Standards (OSHA)

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(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation. (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS (ASTM Standards)

ASTM B633 - Zinc Plating

ASTM D3951 - Packaging Commercial

(Application for copies shall be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION STANDARDS (NEMA Standards)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N. W., Washington D. C. 20037)

SOCIETY OF AUTOMOTIVE ENGINEERS HANDBOOK (SAE Handbook)

SAE J429 - Mechanical and Material Requirements for Externally Threaded Fasteners

SAE J492 - Rivets and Riveting

SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices

SAE J588 - Turn Signal Lamps

SAE J589 - Turn Signal Switch

SAE J843 - Brake System Road Test -- Passenger Car and Light- Duty Truck

SAE J899 - Operator's Seat Dimensions for Off-Road Self-Propelled Work Machines

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SAE J985 - Vision Factors Considerations in Rear View Mirror Design

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096)

AMERICAN NATIONAL STANDARDS INSTITUTE (Current Issues)

ANSI B56.1 - Safety Standard for Powered Industrial Trucks

ANSI B56.8 - American Standard for Personnel and Burden Carriers

(Application for Copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018)

THE TIRE AND RIM ASSOCIATION, INC. YEARBOOK (T & RA Yearbook)

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market St., Akron, OH 44313)

UNDERWRITERS LABORATORIES, INC.

UL583 - Standards for Electric - Battery - Powered Industrial Trucks

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, New York 11756; 207 East Ohio Street, Chicago, Illinois 60611 or 1655 Scott Boulevard, Santa Clara, CA 95050)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first produced unit shall be subjected to a first article inspection (see 6.2) in accordance with 4.4.

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3.2 Materials. Materials not definitely specified shall be at least equal to the end item manufacturer's normal commercial product, and shall meet test requirements specified in Section 4. "Normal commercial product" shall be interpreted to mean an item furnished by an end item manufacturer normally engaged in commercial production of the item(s) covered by this specification.

3.2.1 Castings. Castings shall be of the quality normally used in commercial practice, free from blowholes, porosity, shrinkage, cracks, or other defects. Castings shall not be repaired that adversely affect the strength or intended use. Sufficient finish shall be allowed to permit machining for leak-free sealing at covers, closure points, and fasteners.

3.2.2 Plating. All threaded fasteners, washers and cotter pins required to fabricate the truck, except those in contact with oils in reservoirs and those inside corrosion resistant components shall be zinc plated, cadmium plated or made of corrosion resisting material. Zinc plating shall be in accordance with ASTM B633, type I, SC 1 and cadmium plating shall be in accordance with QQ-P-416, type I, class 3.

3.2.3 Screw threads. Screw threads shall be in accordance with FED-STD- H28.

3.3 Safety. The truck shall conform to the safety standards issued by the Occupational Safety and Health Administration (OSHA) under the Secretary of Labor, and to the applicable requirements in the current issue of UL583 and ANSI B56.8.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories and spare parts.

3.5 Design. The design of the trucks and all components covered by this specification shall provide safe and efficient means for transporting palletized and unpalletized loads of the rated capacity over paved areas and in warehouses. There shall be no evidence of accelerated wear, or failures, or permanent deformation when operated under normal service conditions or when tested as specified in Section 4.

3.5.1 Electrical equipment.

3.5.1.1 Resistors. Resistors shall not be used in series with the motor armature for speed control. Speed control shall be effected by the application of a system as specified in 3.5.2.4.

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Low resistance shunts across field coils may be used to increase top speed of the motor. Such resistors as are necessary in the control circuit network shall be continuous-duty types of a power rating value sufficient to withstand all tests required by this specification.

3.5.1.2 Enclosures and Shields. All trucks shall comply with the requirements of UL 583. All electrical components including wiring shall be located or protected by adequate means to prevent contact by excess moisture or battery electrolyte in the event of spillage.

3.5.1.3 Overcurrent devices. In addition to the requirements of 3.3 overcurrent devices shall be provided to properly protect each power circuit from damage resulting from wiring defects or other faulty operations. The overcurrent devices shall not disconnect power when specified tests, including slope ascension, are conducted with fully charged batteries.

3.5.1.4 Heat Producing Devices, Type EE and EX Trucks. The outside surface temperature of heat-producing devices shall stabilize at temperatures which are allowed by UL 583 while the truck is operating with rated capacity continuously at the speed which creates the most heat.

3.5.1.5 Static discharge device. Type EE and EX trucks shall be equipped with static tires on the steer wheels or two replaceable, electrically conductive straps., mounted adjacent to the drive wheels to discharge static electricity through contact with the road surface. EX trucks shall only have static tires.

3.5.1.6 Control panel. The control panel shall be designed for easy removal as a unit or front removal of electrical assemblies mounted thereon. Mounting studs, bolts, or nuts, shall be so secured that they will not turn or become loose when assembly is removed from the panel. The control panel cover design shall permit unobstructed removal of the cover for inspection and servicing.

3.5.1.7 Conductors. Conductors shall be sized to carry maximum current loads as demanded for heavy duty operations. For type EE trucks all conductors, except battery leads emerging from enclosure(s), shall be through fittings so that sparks shall not propagate to the atmosphere and dust shall not enter from the atmosphere into the enclose(s). Identification of control leads shall be provided through color coding, numbered markings, tags or similar means to identify conductor runs. For type EX trucks, all conductors shall be protected as specified in UL 583.

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3.5.1.8 Fungus and moisture resistance. Unless otherwise specified (see 6.2), the electric circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish with the following exceptions:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. Before varnishing, all surfaces shall be cleaned and prepared so that they are free from all foreign matter which would interfere with the adherence or function of the varnish. The varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of .001 inch (.025mm) to component or element surfaces.

3.5.2 Motor and motor accessories.

3.5.2.1 Motors - Type E trucks. Traction motors shall be industrial type, with a high starting torque, designed for efficient operation on 36 or 48 volts D. C. Motors shall withstand the stresses and current loads induced by test as required by this specification. The resultant strains and temperature rise shall not cause failure of any component part or excessive pitting other than normal discoloration of the commutators. The hydraulic pump and steering pump motor, if used, shall successfully withstand the stresses and current loads induced by tests required by this specification. All electric motors shall have bolt-on conductor connections.

3.5.2.2 Motors - Type EE and EX trucks. In addition to the requirements of 3.5.2.1, motors for type EE and EX trucks shall be provided with one or more thermal relay switches which cause disconnection of battery power to the affected circuit before the outside surface temperature of the motor(s) exceed limits as specified in 3.5.1.4. Thermal relays shall be located in critical heat areas and shall interrupt the current only to the over-heated circuit.

3.5.2.3 Insulation. Insulation for motors shall be Class B, F, or H as defined in NEMA Standards, but the insulation is not to contain silicone materials. Silicone material in any form shall not be used in the motor.

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The electrical and mechanical properties of the insulated windings shall not be impaired by the application of the temperature permitted for the class insulating used (the word "impaired" is here used in the sense of causing any change which could disqualify the insulating material for continuous service).

3.5.2.4 Speed controller. The speed controller shall be a solid state type Silicone Controlled Rectifier (SCR), PMC or MOSFET type. The SCR shall provide infinite variation in travel speed from zero to maximum truck speed. The speed controller shall provide smooth acceleration in both forward and reverse, and shall allow for smooth transition when reversing directions with the truck in motion (plugging). Resistors shall not be used for speed control of the truck. Contactors, where applicable, shall have replaceable tips of silver or copper alloy composition.

3.5.2.5 Safety cutout system.

3.5.2.5.1 Type E truck. Provisions shall be made for type E trucks so that all power to the traction motor is interrupted when the operator leaves his seat.

3.5.2.5.2 Type EE and EX trucks. In order to insure that no current is flowing when the connector is being disconnected on type EE and EX trucks, a magnetic contactor shall be installed in one of the two main power lines from the battery connector. The control circuit for the magnetic contactor shall be wired with two switches in series, either of which shall cause the contactor to open and disconnect all power. The first switch shall be a seat switch. The second switch shall be an emergency switch with a positively attached mushroom type button located within easy reach of the operator. For type EX trucks the emergency switch shall be an enclosure to comply with class I group D and class II group G of UL 583.

3.5.3 Battery and battery accessories. Unless otherwise specified (see 6.2), battery size and capacity shall be as standard with the manufacturer. Battery leads for type E and EE trucks shall be equipped with a two-contact, type EC locking half connector, that is used to connect with electric motor through mounting half or to connect to battery charger with charging half. The battery, battery leads and connectors on type EE and type EX trucks shall be shielded or modified as necessary to assure that the battery as a component of the truck, when drawing maximum current shall comply with 3.9.1 as applicable. When truck is equipped with built in battery charger, EC type connectors are not required.

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When a first produced truck is required (see 6.2), the battery for the first produced truck used for testing by the contractor shall be shipped in a wet and charged condition after the exterior of the battery has been rehabilitated to a "like new" appearance.

3.5.3.1 Battery connector mounting. Type E and type EE trucks shall be equipped with standard EC mounting half connectors, suitably located to fit the locking half as specified in 3.5.3. For type EE trucks, provision shall be made for locking coupled connector with a padlock to be furnished by the Government at destination. Type EX trucks shall have a Type N three contact plug on leads to mate with the charging receptacle specified in MS-16974.

3.5.3.2 Battery compartment. The battery compartment shall be located within the planned outline of the truck. The compartment shall be constructed to permit quick inspection and servicing of battery using simple hand tools and shall provide for easy battery removal. The inside of the compartment shall be finished with one coat of acid and alkaline resistant paint to resist deterioration through normal contact with electrolyte. Compartment construction shall provide for support to give a minimum of 1/2 inch (1.3 cm) clearance between the battery and truck.

3.5.4 Drive unit. Each gear (with the exception of worm gears), sprocket, shaft, and axle, shall be of heat treated alloy steel. Drive gear housing(s) shall be equipped with filler and magnetic drain plugs. All required drain plugs, check plugs, or oil level checking and filling devices in the drive system shall be readily accessible without removing major components. Provisioning shall be made for adjustment where necessary to compensate for wear in the differential and pinion.

3.6 Construction.

3.6.1 Chassis and frame. The frame and related structure shall be fabricated from structural steel and steel castings, and shall withstand 200 percent of rated load without permanent deformation. Frame members, bracing, and all their joints shall provide a rigid unit structure. The frame and related structure shall have sufficient depth to afford protection for working parts and shall withstand, without permanent deformation stresses induced by capacity loads in accordance with the applicable test section 4. The frame and related structures shall have an adequate suspension system for reducing load shocks encountered during normal operation.

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3.6.1.1 Deck . The deck shall be of sufficient thickness to support the rated loads and not bend, buckle, twist or warp, and shall be made from non-skid steel floor plate. The deck plate shall be welded or bolted to all supporting frame members and shall have no sharp edges. It shall be skip-welded to the intermediate frame members. If bolted, bolt holes shall be countersunk and bolt heads shall not protrude above the deck surface. The deck plate and all supporting frame members shall form a rigid level platform. The overall usable deck space shall be as specified in 6.2. Deck height shall be between 12" (304 mm) and 30" (760 mm) high as measured from ground level. If specified (see 6.2) the truck shall have stake pockets and side racks.

3.6.1.2 Stake Pockets. When specified (see 6.2) there shall be 10 stake pockets made of steel tubing. Each stake pocket shall be formed to provide a stop and rigid support for a rack leg. There shall be 4 pockets on each side and 2 at the rear of the platform. Stake pockets shall be welded to the deck.

3.6.1.3 Side Racks. When specified (see 6.2), removable side racks, with tubular steel frame and 1 inch diamond wire mesh (not less than 10W and M gauge) welded to the frame, shall be provided. The racks shall be 26 inches high and extend the entire length of the platform. The side rack legs shall be spaced so that they are interchangeable.

3.6.2 Bearings. Rotating parts shall be mounted on ball or tapered roller bearings, except as specifically indicated herein. Bearing load capacity, including applicable safety and life factors, as recommended by the bearing manufacturer, shall be equal to or greater than maximum bearing loading under normal truck operation.

3.6.3 Wheels. The truck shall be mounted on steel, malleable iron, or cast iron wheels, equipped with tapered roller bearings. Wheel hub bearings shall provide for adjustment for proper running tolerances and shall be sealed with a retainer to prevent leakage of lubricant. Front steer wheels shall track true in accordance with automotive practices, as determined by their respective radii. Wheels, equipped with new tires, shall clear any part of the truck structure by at least 1/4 inch (6 mm) under all conditions of operation.

3.6.3.1 Tires. Tires shall be new and unused and shall be non-directional, pneumatic, high pressure type. Tire loadings shall not exceed the values listed in THE TIRE AND RIM ASSOCIATION Yearbook. Tubeless type tires are permissible.

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3.6.3.2 Tubes. Tubes, when used, shall be heavy duty type, the size required to fit the tires furnished. Valve stems shall be arranged to permit inflation of mounted tires from the side of the truck without removing wheels.

3.6.4 Wheel mounting. Wheels and axles shall be arranged to permit the truck to negotiate the obstacle course in Test Method No. 10, with no more than one displaced wheel off the ground at any time. Means shall be provided to prevent wheels or tires from contacting any part of truck structure, irrespective of obstacle magnitude.

3.6.5 Steering. An automotive type steering wheel, mounted on the steering column, shall turn the front wheels. Clockwise rotation of the steering wheel shall provide for right hand turning of the truck wheel in forward motion. Provision shall be made for precision adjustment to compensate for wear of all major steering parts and for lubrication of all friction points by accessible grease fittings or a lubricant reservoir. When provided the reservoir shall be accessible for filling or checking of lubricant level. Fifth wheel steer will not be acceptable. There shall be a minimum of 7 inches (17.8 cm) measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point on the unoccupied seat cushion.

3.6.6 Body. The truck body shall be constructed of heavy steel sheet. Overall width shall be such that the truck can perform the applicable tests in the Appendix of this specification. Provisions shall be made for immediate and easy access to all working parts requiring inspection or maintenance. Covers shall be secured with bolts or quick opening fasteners removable with ordinary hand tools.

3.6.6.1 Operator's compartment. The operator's compartment shall be designed to accommodate a large sized, seated, clothed male as specified by SAE J899.

3.6.6.2 Seat. A cushioned seat and cushioned backrest shall provide comfortable seating for the operator when operating the truck and shall be in accordance with SAE J899. Provisions shall be made to interrupt all power to the traction motor when the operator leaves the seat.

3.6.6.3 Slinging and tying down attachments. The trucks shall be equipped with lifting and tying down attachments conforming to MIL-STD-209, Type III. Attachments shall be indicated by a transportation plate conforming to MIL-P-514 except tying down attachments may be identified by stencilling or other suitable marking on the truck.

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Tying down markings shall clearly indicate that the attachments are intended for tying down the truck on the carrier when shipped.

3.7 Controls and instrumentation. Where applicable, the provisions of UL 583 shall apply. Electric controls and instrumentation shall be constructed to be moisture and weather resistant. On type "EE" and "EX" trucks, all instruments shall be of spark-enclosed construction or be mounted in a spark-proof enclosure.

3.7.1 Controls. All switch and lever type controls shall be provided with position markings (decalcomania are not acceptable). Lever controls shall be equipped with hand grip knobs. All controls shall be easy to operate, positive in action, and shall show no evidence of deformation.

3.7.1.1 Light switch. An "on-off" light switch shall be located on the instrument panel for operation of vehicle lights.

3.7.1.2 Brake pedal. The brake pedal shall be conveniently located for right foot operation within a clear space at least 6 inches (15.2 cm) wide. Pedal width shall be 3 inches (7.6 cm) minimum. Suitable means shall be provided for lubricating moving parts of the brake pedal and shaft. Required brake performance shall be attained with a pedal force of not more than 125 lb. (57 kg).

3.7.1.3 Accelerator control. The accelerator control shall be installed for comfortable right-foot operation . It shall be located to the right of, and shall be free of interference from, the brake pedal by a distance of not less than 1-1/2 inches (3.8 cm).

3.7.1.4 Directional controls. Selective forward and reverse control lever(s) shall be provided and shall be conveniently located for right hand operation. Forward and reverse motion, however, shall be caused only by actuation of the accelerator pedal.

3.7.2 Instruments. All gages and meters shall be products of manufacturers regularly engaged in producing these types of instruments. They shall be flush in locations convenient to and plainly visible to the operator. All instruments shall be constructed to be moisture and weather resistant.

3.7.2.1 Hour meter. An electrical operational hour meter, which registers the number of operating hours, shall be provided. The totalizing mechanism shall register not less than 9999.

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Hour meters for type 'E' and 'EE' trucks shall be in accordance with MIL-M-3971, Type I, Grade A. Hour meters for 'EX' trucks shall meet UL 583 requirements. The meter shall be weather resistant.

3.7.2.2 Battery Protector. Each unit shall be provided with a discharge indicator, of solid state construction, requiring no adjustments, shall be tamper-proof, weather and water resistant and shall carry Underwriters Laboratories (UL) or Factory Mutual (FM) approval. Recharging or replacing the battery with one fully charged shall be the only means of resetting the warning lights and alarm.

3.8 Accessories. Where applicable, the provisions of UL 583 shall apply. All accessories shall be constructed to be moisture and weather resistant.

3.8.1 Headlights. Unless otherwise specified (see 6.2), each truck shall be equipped with 2 automotive type sealed beam headlights, of not less than 35 watts each.

Headlights shall be recessed behind the front bumper plate to prevent damage without obstructing the light beam. The headlights shall be vertically and horizontally adjustable.

3.8.2 Stoplight. One or more automotive, red, reflector type, combination stoplight and tail lamp, of not less than 15 candlepower and 3 candlepower respectively, shall be recessed or mounted on the rear end within the plan outline of the truck. The stoplight shall operate automatically only upon actuation of the service brake pedal and shall not operate when the parking brake is applied.

3.8.3 Horn. Truck shall be equipped with an electric horn with push button mounted on the center of the steering wheel or steering column. Horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant.

3.8.4 Closed cab. When specified (see 6.2), the truck shall be equipped with a removable rigid closed cab. The cab shall include wind-shield, rear and side windows made of safety glass and shall provide maximum visibility. An electrically operated wiper(s), a windshield defrosting unit and a cab heater shall also be provided.

3.8.5 Other Options When specified (see 6.2), the truck shall be equipped with fore and aft turn signal lamps and switch conforming to SAE J588 and SAE J589, a rearview mirror conforming to SAE J985 (when a closed cab is required), or side mirrors.

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3.9 Performance.

3.9.1 Electromagnetic interference control. Shorebased trucks shall comply with SAE J 551. When specified (see 6.2), shipboard trucks shall be electromagnetically suppressed in accordance with class C1, Group 2, Part 8 of MIL-STD-461 (see 4.6.1).

3.9.2 Steering. The steer wheels shall be turned from extreme right to extreme left or vice versa in not more than 6 complete turns of the hand steering wheel. The force required to operate within the first half of the inner steer angle (from a straight ahead position) shall not exceed 40 pounds (177 Newtons), nor shall it exceed 80 pounds (355 Newtons) at the extreme ends of the steer in either direction when the loaded truck with two steer wheels is stationary on a smooth dry concrete floor. The hand steering wheel diameter shall not exceed 20 inches (51 cm). Other maneuverability features shall meet test course requirements Test Method No. 10 in the Appendix.
The steering mechanism shall withstand an unbalanced tangential force of 100 pounds (444 Newtons) applied at the rim of the steering wheel without permanently distortion, fracturing or deforming any part of the steering mechanism, and at no time shall the free play be greater than 3 inches (76mm) (see 4.6.2).

3.9.3 Speed. While carrying a rated load on a level surface, truck shall move in the forward and reverse speeds of not less than 6-1/2 miles per hour (10.4 km/hr) (see 4.6.3).

3.9.4 Slope ascension, forward direction. Truck shall ascend a 15 percent grade on a dry concrete surface when carrying rated load. Trucks shall accelerate from a dead stop on this slope with rated load (see 4.6.4).

3.9.4.1 Underclearance. The truck shall have sufficient underclearance to permit operation from one horizontal plane to another, up or down a 15 percent slope, and on the obstacle course described in Test Method No. 10 of the Appendix. Hydraulic fittings, hoses, tubing and linkages shall not extend below the lowest portion of the truck and shall be protected by frame members from striking obstacles.

3.9.5 Brakes.

3.9.5.1 Parking brake. The parking brake shall be an independent mechanical friction brake mounted on the drive motor or propeller shaft rotating with the wheels or a separate mechanical linkage actuating the service brake. The parking brake shall hold the truck with rated load on a 15% slope in forward and reverse direction (see 4.6.4).

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The brake shall be automatically activated when the operator leaves the operator's seat and hold the loaded vehicle on the specified slope. A hand operated parking brake lever with a locking device shall be provided in addition to the seat brake. This manual control shall be located within easy reach of the operator and in a position to permit easy and safe movement on and off the vehicle from either side. Adjustment of parking brake shoes or band shall be accomplished without the removal of any major assembly other than tires and wheels. The adjustment shall provide positive means to prevent loss of adjustment through inadvertent disengagement of adjusting device.

3.9.5.2 Service brakes. The service brakes shall conform to the applicable portions of J843 and under all operating conditions specified herein function without fade,noise, uneven braking or malfunction. The brake system shall be an automotive type, adequate for the size and weight of the truck and load, and shall conform to the requirements of ANSI B56.1 (see 4.6.5).

3.9.6 Towing hook. Unless otherwise specified (see 6.2) the truck shall be provided with an automatic pintle-type hook. The towing hook shall withstand, without physical failure, a towing force induced in accordance with paragraph 4.6.6. It shall automatically connect with trailers conforming to MIL-T-17479 and similar towing equipment.

3.9.7 Turning radius. Trucks shall turn in 90 degree intersecting aisles whose width is 96 inches (244 cm). Truck shall turn in one motion, without backing, and shall meet this requirement in not more than 3 trials (see 4.6.7).

3.9.8 Acceleration. From a standing start the truck when carrying rated load shall accelerate to 6 1/2 mph (10.4 km/hr) in the first 22 feet (6.7 meters) (see 4.6.8).

3.9.9 Reliability. Truck shall complete a minimum of 1200 laps in 100 hours of testing without failure when tested in accordance with 4.6.10. Failure shall be defined in 4.6.10. If failure occurs during the test, the cause of failure shall be corrected and the tests restarted at the first lap.

3.9.10 Maintainability. Provision shall be made for ready adjustment, servicing, or replacement of all electrical assemblies and components, battery, brake and components, wearing parts of steering assembly, tires, wheels and horn without removal of any major parts other than covers. If hand access openings are used on the exterior of the truck, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover, as required.

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Frame members or other obstructions shall not interfere with the servicing or draining of oils and lubricants from any assembly or components. Servicing or draining shall not require the use of flexible connections or other special device. Drain plugs shall provide for complete drainage. Compliance with specific maintenance requirements for the equipment shall be determined in accordance with 4.6.11.

3.9.10.1 Maintenance operations. Unless otherwise specified herein, each operation in the following maintenance operations list shall be accomplished by one man in not more than the time specified, using common tools and special tools, if any, furnhished with the truck.

Maintenence Operation (Time in Minutes)	E	EE	EX
a. Remove and Replace Drive Motor Brushes	90	90	150
b. Remove and Replace All Contactor Tips as applicable	60	90	150
c. Remove and Replace All Filters, Screens and Strainers in Hydraulic System as applicable	105	105	105
d. Bleed and Adjust Brakes and Refill Master Cylinder (2 men)	45	45	45

3.10 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubrication. All lubricant reservoirs shall be filled to correct levels with appropriate lubrication.

3.10.1. Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.10.2 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

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3.11 Identification marking. A shipping data plate shall be furnished and installed which shall conform to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material Type I, Grade A Class I material and in addition, shall indicate the silhouette of the truck in transport position indicating the center of gravity, the location and capacity of the lifting and the tiedown attachments and the contract number. The plates shall be attached by screws, drive screws, bolts, or rivets in conspicuous protected locations.

3.11.1 Instruction plates. The truck shall be equipped with instruction plates conforming to MIL-P-514, type III, composition C (GG-P-455, type I, grade A, class 1) material, or commercially proven decals, including warnings, cautions and diagrams describing any special or important procedures to be followed in assembling, operating or servicing the truck.

3.11.2 Wheel loading plate. Each truck shall be equipped with a wheel loading plate conforming to MIL-P-514, Type III, Composition C, (GG-P-455, type I, grade A, class 1) material. Plate shall be attached by screws, bolts, or rivets in a conspicuous location. As a minimum a plate shall have the following information:

Wheel loading (no load)	
Front wheels (each side)	pounds
Rear wheels (each)	pounds
Wheel loading (rated load)	pounds
Front wheels (each side)	pounds
Rear wheels (each)	pounds

3.12 Registration data and vehicle marking. The manufacturer shall execute registration data cards, as required by the contracting officer. Unless otherwise specified (see 6.2), the assigned registration number for each truck shall be painted by the contractor, in black enamel, in block numbers 3 inches on each side of the truck. In addition, the gross vehicle weight (GVW) with the required battery shall be marked on each side of the truck. The letter shall be gothic capitals, and the numerals shall be 2 inches high where space permits. Paint shall be as specified in paragraph 3.12.1.

3.12.1 Special marking. Trucks shall be marked in black color number 17038 in accordance with FED-STD-595. Characters shall be block-type capitals and Arabic numerals. Marking location and information and character size shall be as follows:

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Information	Example	Location	Character Size
Capacity	4000 Lbs.	On each side	2 inch height
Rider warning	NO RIDERS	On rear	2 inch height
Service designation	U.S. NAVY	On each side	1-1/2 inch ht.
and Registration number	14-XXXXX	and rear	

Durable markers indicating the applicable safety designation of truck shall be applied to each side of the vehicle in a visible but protected location.

3.13 Treatment and painting. All exterior surfaces of the truck shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt and rust and shall be painted as soon as practicable after cleaning with not,less than one coat of primer and two coats of synthetic enamel. Primer and finish coats shall be applied to assure complete coverage and durability of the finish. The finish color for the type E and EE trucks shall be color 13538 of FED-STD-595. The finish color for the type EX truck shall also be 13538, together with a single 6-inch (150 mm) wide blue diagonal stripe at 45 degrees across the right, left, and rear of truck; in addition, the type EX truck shall be identified as to type by painting 4-inch (100 mm) high black block letters with "EX" in a conspicuous space at right, left, and rear of truck. When a first produced truck is required for test and evaluation, it shall be painted with one primer coat only. The finish coat, when dry, shall be a smooth even surface, free from runs, sags, peels, chips, blisters, areas of thin film or no film. Surfaces of components and assemblies of the truck not normally painted with a finish coat shall be cleaned and prepared in accordance with good commercial practice. Paint must be kept off working surfaces where interference with working of parts would result. The paint shall be lead free and chromate free. In addition, the truck shall be stenciled with the words "Painted with lead free, chromate free paint on (date) by (contractor's name)."

3.14 Repair parts and maintenance tools. When specified, specially designed maintenance tools shall be furnished by the contractor (see 6.2).

3.15 Workmanship. The truck shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, spurs, scale, flux, and other harmful extraneous material.

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Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.15.1 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.15.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the forklift to conform to the examination and test requirements specified in Section 4. Parts to be joined by filler welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.187 inch (5 mm) unless appropriate bridging techniques are used. The welding process used in fabrication of the forklift shall be at the option of the supplier.

3.15.3 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers, lockwashers or other positive locking devices shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.15.4 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having corroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components from re-entrant cuts. Heated metals shall be allowed to cool slowly, except for designed heat treatment, and over-heating shall be avoided in accordance with the recommendations of the metal manufacturer. All modular assembly fabrication shall provide for interchangeability of components.

3.15.5 Machine work. Tolerances and gauges for metal fits shall conform to the limits specified herein.

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3.15.6 Castings and forgings. All parts, components, and assemblies of the forklift which include castings and forgings shall be clean of extraneous material such as sand, dirt, sprues, scale and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function. Castings with fastener holes shall provide for correct seating of fasteners such as bolt, lockwasher and nut assemblies and capscrews.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections.

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First Article Inspection. First Article inspection shall be performed on one complete truck when a first article sample is required (see 3.1 and 6.2). This inspection shall include the examination of 4.5 and the tests marked "X" in the first column of Table II.

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4.4 Quality Conformance Inspection. Quality conformance inspection shall include the examination of 4.5 and the tests marked "X" in the second column of Table II.

4.5 Examination. First article examination shall include those items marked "X" in the first column of Table I. All cranes shall be examined as marked "X" in the second column of Table I. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5.1 Pretest examination Prior to testing in accordance with 4.6., the truck(s) shall be examined for the defects as specified in Table I, Column I.

4.6 Tests. Upon successful completion of the inspection, the truck(s) shall be subjected to tests in accordance with Table II, Column 1. All tests shall be conducted in any order desired, except that the reliablility and post test shall be conducted last. Acceptance of the first produced truck(s) shall not exclude the remaining trucks from the quality conformance inspections and tests specified in Section 4.

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TABLE I. Examination Schedule

First Produced	Quality Conformance	Defect	Requirement Paragraph
x		Material not as specified	3.2
x	x	Casting not as specified	3.2.1
x		Plating not as specified	3.2.2
x		Screw threads not as specified	3.2.3
x		Safety not as specified	3.3
x	x	Interchangeability not as specified	3.4
x	x	Design not as specified	3.5
x		Resistors not as specified	3.5.1.1
x	x	Enclosures and shields not as specified	3.5.1.2
x	x	Overcurrent devices not as specified	3.5.1.3
x		Heat producing devices not as specified	3.5.1.4
x	x	Static discharge devices not as specified	3.5.1.5
x		Control panel not as specified	3.5.1.6
x		Conductors not as specified	3.5.1.7
x		Fungus and moisture resistance not as specified	3.5.1.8
x	x	Motors and motor	3.5.2.1
		connections not as specified	3.5.2.2
x		Insulation not as specified	3.5.2.3
x		Speed controller not as specified	3.5.2.4
x		Safety cutout system not as specified	3.5.2.5
x	x	Battery not as specified	3.5.3
x		Battery connector mounting not as specified	3.5.3.1
x	x	Battery compartment not as specified	3.5.3.2
x		Drive unit not as specified	3.5.4
x	x	Chassis and frame not as specified	3.6.1
x	x	Deck not as specified	3.6.1.1
x	x	Stake pockets not as specified	3.6.1.2
x	x	Side racks not as specified	3.6.1.3

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TABLE I. Examination Schedule(con't)

First Produced	Quality Conformance	Defect	Requirement Paragraph
x		Bearings not as specified	3.6.2
x		Wheels not as specified	3.6.3
x		Tires not as specified	3.6.3.1
x		Tubes not as specified	3.6.3.2
x		Wheel mounting not as specified	3.6.4
x		Steering not as specified	3.6.5
x		Truck body not as specified	3.6.6
x	x	Operators compartment not as specified	3.6.6.1
x	x	Seat not as specified	3.6.6.2
x	x	Slinging and tiedown attachments not as specified	3.6.6.3
x	x	Controls and instrumentation not as specified	3.7
x	x	Battery protector not as specified	3.7.2.2
x	x	Headlights not as specified	3.8.1
x	x	Stoplight not as specified	3.8.2
x	x	Horn not as specified	3.8.3
x	x	Lubrication fittings not as specified	3.10.1
x	x	Nameplate not as specified	3.11
x	x	Registration data plate not as specified	3.12
x	x	Treatment and painting not as specified	3.13
x	x	Maintenance tools not as specified	3.14
x	x	Workmanship not as specified	3.15

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TABLE II. Test Schedule

First Prod.	Quality Conform.	Post Test	Test	Test Para.	Requir. Para.	Test Method No.
x			Electromagnetic interference	4.6.1	3.9.1	
x		x	Steering wheel test	4.6.2	3.9.2	1
x		x	Wear of steer mechanism	4.6.2	3.9.2	2
x	x	x	Speed	4.6.3	3.9.3	3
x	x	x	Slope ascension Underclearance Parking brake	4.6.4	3.9.4 3.9.4.1 3.9.5.1	4
x	x	x	Service brakes	4.6.5	3.9.5.2	5
x			Towing hook	4.6.6	3.9.6	6
x			Turning radius	4.6.7	3.9.7	7
x		x	Acceleration	4.6.8	3.9.8	8
x			Overload test	4.6.9	3.6.1	9
x			Reliability test	4.6.10	3.9.9	10
x			Maintainability	4.6.11	3.9.10	11

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4.6.1 Electromagnetic interference. Test to determine compliance with 3.9.7 shall be performed in accordance with the requirements of SAE J551. Failure to meet requirements will be cause for rejection.

4.6.2 Steering. Test truck in accordance with Test Methods No. 1 and 2. Nonconformance with the requirements of 3.9.2 shall constitute failure of the test.

4.6.3 Speed. Test the truck in accordance with Test Method No. 3. Noncompliance with 3.9.3 shall constitute failure of the test.

4.6.4 Slope ascension. Parking brake. Underclearance. Test the truck in accordance with Test Method No. 4. Nonconformance to 3.9.4, 3.9.5.1 or 3.9.5 shall constitute failure of this test.

4.6.5 Service brakes. Test the truck in accordance with Test Method No. 5 and 6. Nonconformance with 3.9.5.2 shall constitute failure of this test.

4.6.6 Towing hook. Test truck in accordance with the applicable part of Test Method No. 6. Nonconformance with the requirements of 3.9.6 shall constitute failure of this part of the test.

4.6.7 Turning radius. Test truck in accordance with Test Method No. 7. Failure to conform with the requirements of 3.9.7 shall constitute failure of the test.

4.6.8 Acceleration. Test the truck in accordance with Test Method No. 8. Nonconformance to 3.9.6 shall constitute failure of this test.

4.6.9 Overload. Test the truck in accordance with Test Method No. 9. Nonconformance with the requirements of 3.6.1 shall constitute failure of the test.

4.6.10 Reliability. Test the truck in accordance with Test Method No. 10. Failure for the purpose of the test is defined as any malfunction which cannot be remedied by adjustment, repair or replacement by one person using the organizations tools and parts within 30 minutes and which may cause:

a. Failure to start operation, stop of operation, or degradation of performance capabilities of the truck below designated levels.

b. Serious damage to the truck by continuous operation.

c. Serious personnel safety hazard.

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4.6.11 Maintainability. Test the truck in accordance with Test Method No. 11. Nonconformance to 3.9.10 and 3.9.10.1 shall constitute failure of this test.

4.6.12 Post test disassembly and inspection. Upon completion of tests specified in 4.6 the truck shall be subjected to the tests in accordance with Table II, Column 3. Nonconformance with applicable requirements shall constitute failure. Upon completion of all tests dismantle the truck sufficiently to permit detailed disassembly and visual inspection of the following components.

a. Motors

b. Differential

c. Rear axle bearing and seals

d. Complete brake system components

e. Steering system cylinders, valve and pump (if used).

4.6.13 Failure. Inability to meet or noncompliance with any of the test requirements will constitute failure. Each test failed must be restarted at the beginning of that test. Modification of any components on the truck during testing shall be cause for restarting the entire first article testing.

4.7 Quality conformance inspection.

4.7.1 Test. Each truck shall be subjected to the tests in accordance with Table II, column 2.
Failure of any tests shall be cause for performing the inspection specified in 4.4.2.14.

4.7.2 Inspection. After successful completion of all tests specified in 4.5.1, each truck shall be examined in accordance with Table I, column 2. Presence of one or more defects shall be cause for rejection.

4.8 Inspection comparison. The Government may select trucks at any time during the contract production period to be inspected as specified in column 1 of Table I or column 1 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, and will not include the previously inspected first produced truck(s). In addition to any test specified as part of the inspection comparison, the government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison.

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Failure of such additional tests shall have the same effect as failure of the tests specified as inspection comparison.

4.8.1 Inspection failure. Failure of an inspection comparison truck to meet any requirement specified herein during and as a result of the inspection and tests specified in 4.6 shall be cause for rejection of the inspection comparison truck(s) and shall be cause for refusal by the Government to continue acceptance of the production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of the deficiencies shall be accomplished by the supplier at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all trucks accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.9 Inspection of preparation for delivery. An inspection will be made to determine that the requirements for preservation, packaging, packing and marking are in compliance with MIL-STD-162.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be in accordance with MIL-STD-162 or ASTM D3951 as specified (see 6.2).

6. NOTES

6.1 Intended use. The heavy duty, electric powered, sit-down rider truck with fixed platform is intended for transporting cargo in and around warehouses, airfields, loading platforms and docks within the military supply system. Trucks are intended for operation over paved and semi-improved surfaces.

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6.2 Acquisition requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of truck required (see 1.2.1).

(c) Capacity of truck required (see 1.2.2)

(d) Deck size required (see 1.2.3)

(e) When first produced truck(s) is (are) required. (see 3.1 and 4.3).

(f) When the Government will conduct any of all of the first produced tests (see 3.1).

(g) When plating other than required is specified (see 3.2.2)

(h) When battery size and capacity are not according to manufacturer's standard size. (see 3.5.3)

(i) When deck holes, stake pockets, and side racks are specified (see 3.6.1.2 and 3.6.1.3).

(j) When towing hook is required (see 3.9.6)

(k) When two headlights are required (see 3.8.1)

(l) When rigid closed cab is required (see 3.8.4)

(m) When other options are specified (see 3.8.5)

(n) When electromagnetic interference control is required (see 3.9.1)

(o) When fungus and moisture resistance are not required (see 3.5.1.8)

(p) When registration data is other than that required (see 3.12)

(q) When maintenance tools are required (see 3.14)

(r) Special markings if required in addition to MIL-STD-129 (see 3.12.1)

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6.3 First article. When a first article inspection is required, the item(s) should be the first produced item(s). The first article should consist of one or two units. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article(s). Invitations requirement for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Conditions for use of level B preservation. When level B preservation is specified (see 5.1), this level of protection should be reserved for the acquisition of trucks for resupply worldwide under known favorable handling, transportation and storage conditions.

6.5 Subject term (key word) listing.

Truck, fixed platform
Brushes, drive motor
Button, mushroom type
Pockets, stake
Hook, pintle-type
Underclearance

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX

FOR

TRUCK, FIXED PLATFORM, ELECTRIC POWERED

10. SCOPE

The test methods contained within this appendix are to determine whether trucks procured under this specification conform to the requirements set forth. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. TEST METHODS.

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TEST METHOD NO. 1

STEERING WHEEL TEST

Purpose: To determine whether the force required to turn the steering wheel meets specification.

Test Description: Place the unloaded truck on a level concrete surface. Using a tension dynamometer, or equivalent, measure the force required to move the steering wheel.

Test Procedure:

1. Place truck on dry, level, smooth, trowelled or brushed concrete with steer wheel(s) in a straight-ahead position.

2. Assemble an adapter pulley having a root diameter equal to the steering wheel diameter and equipped with means for attachment to the wheel spokes.

3. Attach a tension dynamometer to the adapter pulley.

4. Grasp the dynamometer lead and apply a steady force, in a counter-clockwise direction and in the wheel plane, until the wheel turns.

5. Observe the maximum gage reading.

6. Perform 4 and 5 in a clockwise direction.

7. Turn steering wheel clockwise until the steer wheels hit the stop. Apply a 150 lb. unbalanced tangential force to the steering wheel and maintain the force for a minimum of 15 seconds. Perform this operation two more times.

8. Repeat step 7 in the counter-clockwise direction.

9. Measure the number of turns of the hand steering wheel required to turn the steer wheels from the extreme right position to the extreme left position. Repeat this operation in the opposite direction.

10. Measure the steering wheel diameter, and the steering wheel to seat clearance.

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TEST METHOD NO. 2

WEAR OF STEERING MECHANISM *

Purpose: To determine cumulative wear in the steering mechanism of the truck due to carrying out requirements of the reliability test.

Test Description: With the truck stationary and with out load, the amount of free play in the steering wheel is measured.

Test Procedure:

1. Place unloaded truck on a dry, level, concrete surface with its front wheels parallel to the longitudinal axis of the truck.

2. Attach to the steering column, immediately below the steering wheel and parallel to it, a piece of stiff fiber board or similar material which is semi-circular in shape and which has a radius of at least one inch greater than the radius of the hand steering wheel.

3. Mark a reference point on the upper portion of the outer edge of the hand steering wheel; mark a corresponding point on the fiber board templet.

4. Turn the steering wheel clockwise until the front wheels of the truck begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.

5. Turn the steering wheel clockwise until the front wheels of the truck begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.

6. Remove fiber board templet from truck.

7. Place fiber board templet on a flat surface and measure the distance between two points made in accordance with operation 4 and 5 above. This distance, in inches, is the free play of the steering mechanism.

*This test should be done before and after reliability test.

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TEST METHOD NO. 3

SPEED

Purpose: To determine whether truck meets the speed requirements of this specification

Test Description: Measuring the speed of the truck, with rated load, a distance of at lease 88 feet (27 meters).

Test Procedure:

1. Drive the truck a sufficient distance to attain maximum speed prior to entering the measured test course.

2. Record the length of measured distance and time to traverse the measured distance.

3. Repeat for a total of six runs, in forward gear, three in each direction.

4. Repeat a total of six runs in reverse gear, three in each direction. Travel speeds in forward and reverse shall be the average of the six runs.

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TEST METHOD NO. 4

SLOPE ASCENSION - PARKING BRAKE - UNDERCLEARANCE

Purpose: To determine whether truck, when carrying rated load, can accelerate from a dead stop on a 15 percent grade and has sufficient underclearance to operate from a 15 percent slope to a horizontal surface. To determine whether the parking brake can hold truck with rated load on a 15 percent slope in both forward and reverse direction.

Test Description: Truck ascends a 15 percent grade in a forward direction on a dry concrete surface. Truck stops and starts up slope again. Horizontal approaches should be provided for the ramp.

Test Procedure:

1. Place rated load on platform.

2. Drive truck along horizontal approach to the 15 percent ramp.

3. Drive truck approximately half-way up the 15 percent ramp and bring to a complete stop.

4. Apply parking brake, release service brakes and observe whether truck remains stationary.

5. Release parking brake, and start truck from dead start and proceed up ramp and onto a horizontal surface. Observe underclearance.

6. Descend ramp. Stop on ramp and apply parking brakes, release service brakes, and observe whether truck remains stationary.

7. Proceed on to the horizontal surface.

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TEST METHOD NO. 5

SERVICE BRAKE PRESSURE

Purpose: To determine whether brake pedal and brake system have the strength required by the specifications.

Test Description: Using a pedal force gauge, apply a pressure of 250 pounds (1112 Newtons) for one minute to the brake pedal a total of 10 times.

Test Procedure:

1. Attach the pressure gage to the brake pedal in a manner which enables pressure to be applied to the face of the gauge.

2. Apply foot pressure to the gauge until it registers 250 pounds (1112 Newtons) and hold for 1 minute.

3. Repeat operation number two, 10 times.

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TEST METHOD NO. 6

SERVICE BRAKE PRESSURE SENSITIVITY,

TOWING HOOK STRENGTH

Purpose: To determine whether a brake pedal pressure within the range of 75 to 125 pounds (333 to 555 Newtons) will develop a draw bar pull equal to approximately 18 percent of the gross weight of the truck (with rated load) while truck is being pulled.

Test Description: Pull truck with rated load over dry, smooth, trowelled or brushed concrete having a coefficient of friction not less than .50, with parking brake released.

Test Procedure:

1. Determine the gross weight of truck plus rated load.

2. Attach pressure gage to brake pedal, in a manner which will enable pressure to be applied to the face of the gage.

3. Attach a tension dynamometer to the front of the truck.

4. Attach one end of a block and tackle, or similar device for securing mechanical advantage, to the tension dynamometer, and the other end to an anchoring device.

5. Apply sufficient brake pedal pressure to prevent wheel from turning.

6. Apply tension to the dynamometer until force is approximately equal to 30 percent of the total weight of truck plus rated load.

7. Gradually reduce brake pedal pressure until wheels just begin to turn while maintaining the dynamometer tension. This pedal pressure shall not be less than 75 pounds (333 Newtons) no more than 125 pounds (533 Newtons).

8. Repeat operations 2 through 7 with dynamometer attached to rear towing hook.

9. Examine towing hook for evidence of permanent deformation.

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TEST METHOD NO. 7

TURNING RADIUS

Purpose: To determine whether truck can turn in 90 degree intersecting aisle of specified width.

Test Description : 90 degree intersecting aisles of specified width are laid out. The truck must make a right hand turn within the specified aisle widths in one motion, without backing, and with no more than three trials. The test is repeated with the truck making a left hand turn.

Test Procedure:

1. Layout on dry, smooth, trowelled or brushed concrete two 90 degree intersecting aisles of specified width.

2. Drive truck at slow speed, in one motion and without backing, around the intersecting aisles to the right. Truck must negotiate the turn within the specified aisle width in not more than three trials.

3. Perform operation 2 with truck turning to the left.

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TEST METHOD NO. 8

ACCELERATION

Purpose: To determine acceleration of truck with load on a level surface from a standing start.

Test Description: Truck is accelerated at maximum rate from a standing start, timed through a distance of 22 feet (6.7 meters), and then brought to a stop.

Course Layout:

A----------22' (6.7 m)----------B

Test Procedure:

1. Accelerate the truck in forward gear at maximum speed through the specified distance from a standing stop.

2. Repeat a total of six runs, three in each direction.

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TEST METHOD NO. 9

OVERLOAD TEST

Purpose: To determine whether the truck structure can withstand specified overload without permanent visible deflection or deformation.

Test Description: With truck on a level surface, specified overload shall be placed on the truck platform. There shall be no permanent deformation or failures of related structures.

Test Procedure:

1. Place truck on level surface.

2. With crane or forklift, the truck shall be uniformly loaded with specified overload. When placing the overload on trucks, personnel should stand clear and loads should be removed if there is evidence of immediate collapse or breakage in truck structure.

3. If truck accepts loads without immediate failure, let overloads stand to 10 minutes on truck.

4. Remove loads.

5. Inspect truck structure and frame for evidence of deformation ,fractures, broken welds, etc.

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TEST METHOD NO. 10

RELIABILITY TEST

1. Test Course:

(a) Layout of course. The test course shall be set up in accordance with the provisions of Drawing A. The length of each leg may be varied, so long as the total length of the course and its general proportions are maintained. Aisle or road widths shall be 12 feet (3.6 meters). The perimeter of the course shall be encompassed with suitable barriers or indicators. Rubber pylons, or other suitable markings shall be placed at all corners and other critical points to assist in keeping the machine under test within the course limits.

(b) Ramp inclusion. The course shall be laid out to include the slope and approximate length specified (plus or minus one percent) on Drawing C.

(c) Obstacle inclusion. Obstacles shall be included at point (G) in the course layout as indicated on Drawing (B). Iron, wood, cement or steel blocks may be used. A guideline 15 feet (4.6 meters) long shall be provided to the left of the centerline of the course and parallel to it. This line shall be at a distance from the course centerline equal to one-half the distance between front wheel centers when two steer wheels are used. This spacing and arrangement will allow the left wheels of the machine under test to pass over the first pair of blocks and then the right wheels to pass over the second pair of blocks and while the machine is moving forward in a straight line with the left front wheel tracking on the guideline.

(d) Course surface. The test course shall be concrete or macadam, and shall be free of snow, ice, any non-planned obstacles or foreign materials during the course test.

(e) Minimum lighting requirements. Minimum general illumination of the course shall be 5-foot-candles (5.38 lux) on the operating level. However, lights of the truck shall be on at all times during the test. The light switch shall be turned "off" and "on" at the beginning of each lap.

(f) Changing batteries. Each machine under test shall utilize not more than two sets of fully charged batteries during the course of each operating 8-hour day of the performance test. However, the time required to change batteries under the provisions of this paragraph shall be excluded from the operating time. Batteries used shall have minimum ratings as specified.

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(g) Speed of negotiating course. Each machine undergoing test should traverse each part of the performance test course at the maximum safe practical speed.

(h) Laps per hour and per day. Each truck shall complete a minimum of 1200 complete circuits of the test course in 100 hours of testing. Circuits (laps) shall be conducted on a continuous basis during each day (8 hour minimum).

(i) Duration of the performance test. The performance test specified in this standard shall be conducted on a continuous basis each operating day until the minimum requirements are accomplished.

(j) Periodic maintenance. On the fourth operating day (after 24 hours) of the performance test the truck shall undergo routine maintenance procedures. However, no repairs or replacement of parts shall be permitted.

(k) Method of negotiating course. The machine under test shall begin each lap at the point labeled "START" on Drawing A. It shall have been loaded with rated load prior to start of the test. As the operator starts the machine he shall operate the horn for approximately 1 second; turn on headlights, turn off and turn on. Proceed in a forward direction along the portion of the course marked "A" until it reaches the first turn area marked "B" completing a 360^0 turn in a clockwise direction and continue to stopping point. After stopping, the machine shall make a 90^0 turn and proceed onto the slope or ramp. The machine under test shall proceed up the slope or ramp to the position labeled "Stopping Point Under Load" on the drawing. Here the driver shall stop the machine and then resume mounting the slope or ramp. After reaching the top of the slope or ramp, the machine shall proceed along the main aisle in the portion of the course marked "E" and "F" until it reaches the portion of the course marked "G" which contains the obstacle test. The obstacle test shall be traversed by the machine being driven through this area in a straight line, with its left front wheel on the guide line, so that its left wheels pass over the first pair of blocks and its right wheels pass over the second pair of blocks. This part of the test may be conducted at low speed. However, the machine shall not be stopped prior to going over the blocks. After the obstacle test has been traversed the machine shall proceed along the main aisle in the portion of the course marked "H" until it reaches the final turn and stop. Here the machine shall make a tight 360^0 turn, in a counterclockwise direction at maximum acceleration for the first 90^0 of the turn, land then stop upon completion of the turn.

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The driver shall continue to maneuver the machine around the course, in the manner described above, in accordance with 1. (g). On alternate days (after 8 hours), the course shall be traversed in the opposite (or clockwise) direction. On these days the test procedures shall be reversed. The "Hourly Record Sheet" shall be filled out for each operating hour of the performance test.

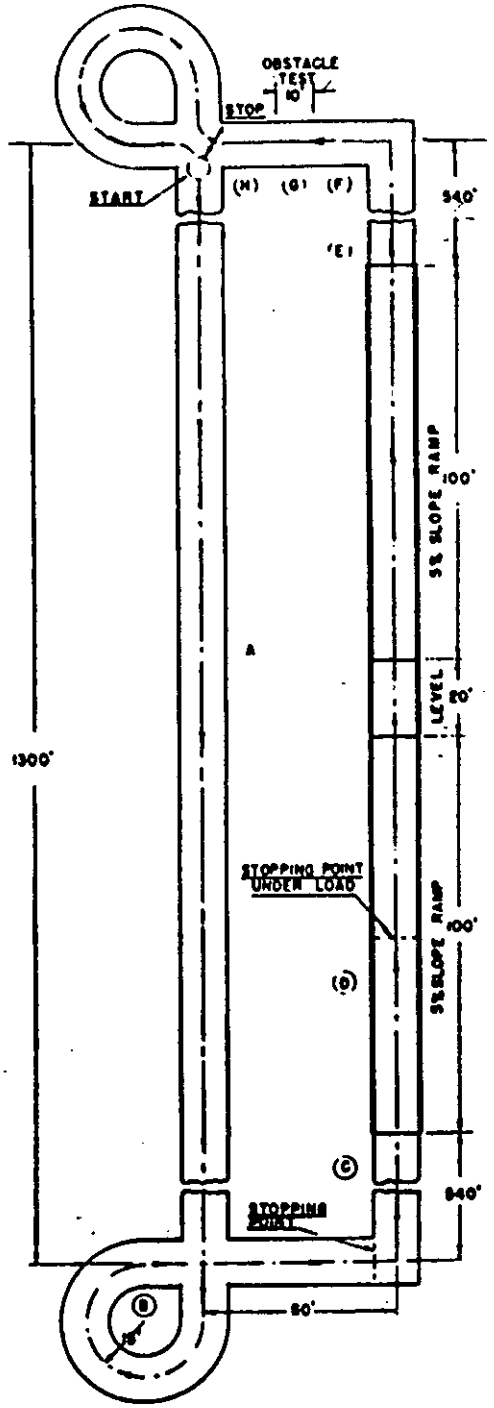
(l) Operating limitations. The truck under test shall not be permitted to leave the bounds circumscribed in Drawing A.

(m) Failure. Any failure of the truck during the performance testing that requires more than thirty minutes to repair (one person) shall be cause to restart the test from the first lap.

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DRAWING "A"

OUTDOOR TEST COURSE FOR PLATFORM TRUCKS

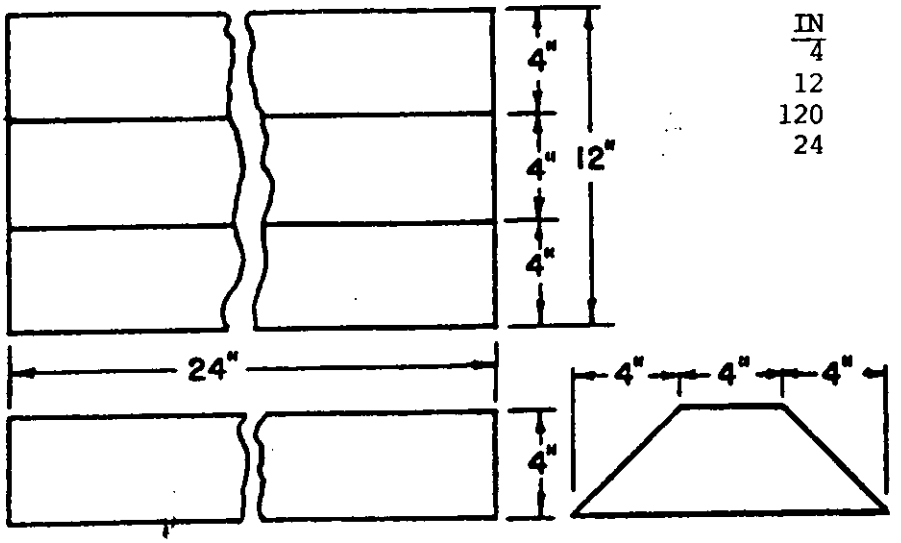


FEET	METERS
10	3
15	4.6
20	6
50	152
100	30
540	164
1300	396

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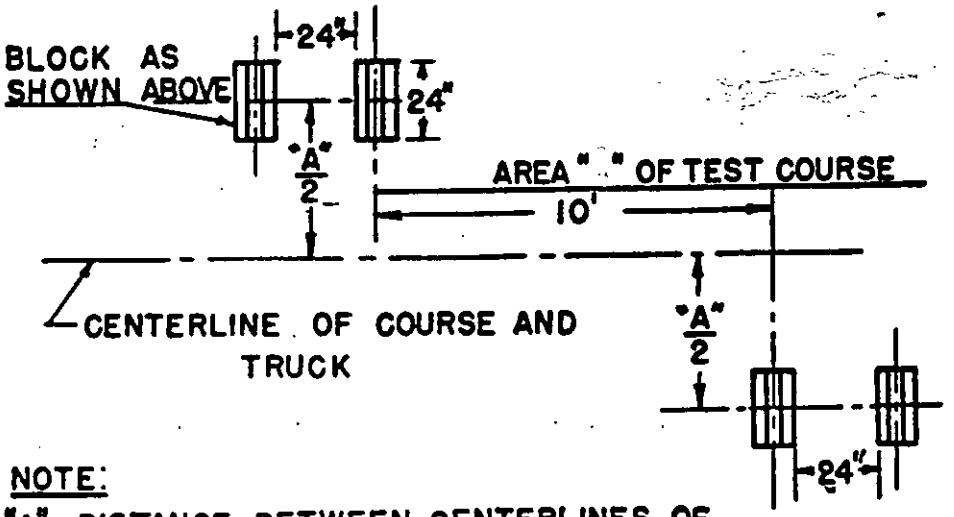
DRAWING "B"

DETAILS OF OBSTACLE BLOCK CONSTRUCTION



IN	CM
4	10
12	30
120	300
24	60

LAYOUT OF OBSTACLE TEST

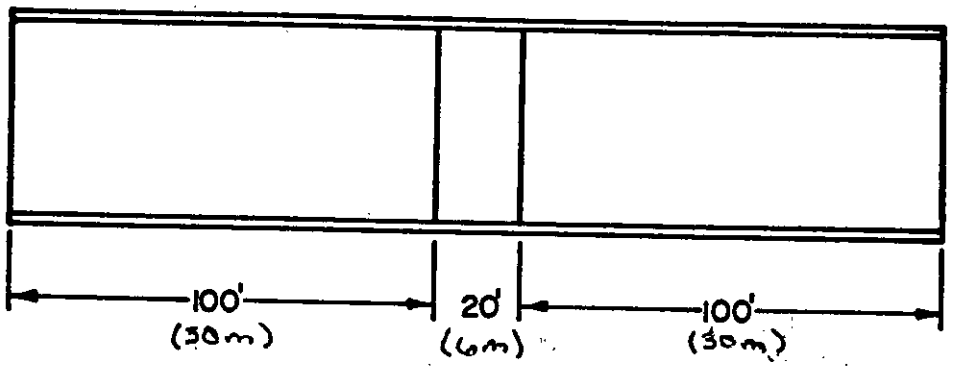


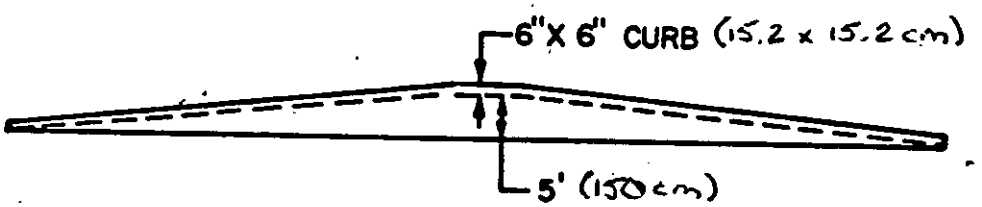
NOTE:

"A" = DISTANCE BETWEEN CENTERLINES OF DRIVING WHEELS

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DRAWING "C"





OUTDOOR TEST RAMP

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HOURLY TIME RECORD

TRUCK MANUFACTURER	MODEL	SERIAL NO.	DATE

TEST LOCATION	RECORDED BY	CLOCK READINGS:	(Start)	(Finish)	REQ'D LAPS/HR

LAP NUMBER	WATCH READING	ELAPSED TIME	REMARKS
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
TOTAL			AVERAGE LAP TIME

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Instructions for Timing Performance Test

1. Start decimal minute stop at zero at beginning of each operating hour.

2. Record watch reading at end of each lap in "Watch Reading" space.

3. Subtract watch reading at end of preceding lap from reading at end of lap. Record difference in "Elapsed Time" space.

4. If equipment under test completes the required number of laps prior to completion of an operating hour, it shall be stopped until start of next hour, and then resume performance test.

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TEST METHOD NO. 11

MAINTAINABILITY

1. Test Course:

Shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. Test Apparatus:

(a) Common handtools

(b) Special tools (if furnished with truck)

3. Test Procedure:

(a) Examine truck for accessibility of major assemblies.

(b) Examine position of drains with respect to accessibility.

3. Test Procedure:

(a) Examine truck for accessibility of major assemblies

(b) Examine position of drains with respect to accessibility.

(c) Using the tools, demonstrate that major assemblies are accessible for repair and maintenance.

(d) Demonstrate both the accessibility of drains and the path of discharge of lubricants by activating drainage controls.

(e) Perform and time each operation on the maintenance operations list (see 3.4.1).

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Custodians:

Army - ME
Navy - SA
Air Force - 99

Review Activities:

Navy - OS
Air Force - 84
DLA - CS

Preparing Activity:

Navy - SA

Project No. 3930-0671

User Activities:

Army - SM, ER
Navy - SH, AS, MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-3997 E	940415

3. DOCUMENT TITLE
TRUCK, FIXED PLATFORM, ELECTRIC POWERED, PNEUMATIC RUBBER

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*
TIRES

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER ATTN: CODE 0341	(1) Commercial (717)790-5508	(2) AUTOVON 430-5508
c. ADDRESS (Include Zip Code) NAVY SHIPS PARTS CONTROL CENTER PO BOX 2020 MECHANICSBURG, PA 17055	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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